

1. Derive each of the convolutions listed in Table 2-2 of your textbook (reproduced below), and submit your handwritten work. You are expected to show your detailed work, following the systematic procedure discussed in class. Make sure to show how you arrived at the integration limits, and the ranges of valid values for time (t) for each derivation. In other words, this exercise is intended to help you achieve a disciplined approach to computing convolution integrals. Item 1 is worth 10 points, and the rest are worth 15 points each.

Table 2-2: Commonly encountered convolutions.

1. $u(t) * u(t) = t u(t)$
2. $e^{at} u(t) * u(t) = \left(\frac{e^{at} - 1}{a} \right) u(t)$
3. $e^{at} u(t) * e^{bt} u(t) = \left[\frac{e^{at} - e^{bt}}{a - b} \right] u(t), \quad a \neq b$
4. $e^{at} u(t) * e^{at} u(t) = t e^{at} u(t)$
5. $t e^{at} u(t) * e^{bt} u(t) = \frac{e^{bt} - e^{at} + (a - b) t e^{at}}{(a - b)^2} u(t),$
 $a \neq b$
6. $t e^{at} u(t) * e^{at} u(t) = \frac{1}{2} t^2 e^{at} u(t)$
7. $\delta(t - T_1) * \delta(t - T_2) = \delta(t - T_1 - T_2)$